



CS-319 Project Presentation

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Lab Coordinator: [Eray Tüzün](#)

Project: [BILSEN Review Process Automation](#)

BILSEN Group



Vision

Enhance the **productivity of software engineering** teams through **data-driven and data science**–based techniques.

Group Composition

- Graduate students: 9
- Undergraduate students: 18



Automating the BILSEN Internal Review Process

Context

- BILSEN conducts internal paper reviews before conference/journal submission.
- Reviews are essential for maintaining research quality and consistency.

Goal

- Transform the current **manual, and coordination-heavy review process** into a **structured, partially automated, and data-driven system**.
- Improve efficiency, traceability, and scalability.



Current Process (As-Is): Step-by-Step Flow

1. Paper Submission

- Authors notify the lab coordinator upon completing the final version of their manuscript.

2. Reviewer Assignment

- Coordinator manually assigns reviewers, sets the review deadline, shares the project Overleaf link with the assigned reviewers and notifies them by email.
- The lab coordinator sometimes needs to remind reviewers of their tasks, as missed deadlines or late extension requests can prevent timely reassignment and lead to missed reviews.

3. Review Execution

- Reviewers:
 - Join the Overleaf and read the manuscript
 - Leave free-form comments on Overleaf



Sample Email Sent to Assigned Reviewers by the Coordinator

BILSEN Paper review - Due Date Jan 22 Thursday 17:00 Inbox x



eraytuzun@cs.bilkent.edu.tr

Jan 19, 2026, 11:08 PM (13 days ago)



to ilke, Atakan, mehmet.dedeler@ug.bilkent.edu.tr, me, Emre, zehra.iyigun@alumni.bilkent.edu.tr, izgi.tamci@alumni.bilkent.edu.tr, begum.kara@alumni.bilkent.edu.tr, aytekin.ismail@bilkent.edu.tr

Dear All,

Can you please review the following paper? The paper title is "LACY: Simulating Expert Mentoring for Software Onboarding with Code Tours"

You can access the paper on overleaf by using the link below (please let us know if you cannot access it).

Overleaf Link: <https://www.overleaf.com/7492737938htkzghgvvtc#741d8e>

Deadline: Jan 22 Thursday, 2026, 17:00. (This is strict deadline. Please mark this deadline as soon as you receive this email)

Please spend at least 3 hours to review a paper. Please don't wait until few hours to the deadline and perform a quick look and pretend that the job is done. Your reviews are very important to the authors.

Please do a Reply-All to me and the authors indicating that you are done with your review. Thank you for your help.

Venue

We will submit this to FSE 2026 Industry track <https://conf.researchr.org/track/ease-2026/ease-2026-research-papers>

Notice that this is not a Research Track paper, but an industry track paper. Please first read the call for papers understand the requirements, and then review.

Sample Reminder Email Sent by the Lab Coordinator



Re: BILSEN Paper review



eraytuzun@cs.bilkent.edu.tr <eraytuzun@cs.bilkent.edu.tr>

2 April 2024 Tuesday at 23:42

To: [REDACTED]

From: eraytuzun@cs.bilkent.edu.tr <eraytuzun@cs.bilkent.edu.tr>

Date: Monday, April 1, 2024 at 21:22

To: [REDACTED]

Subject: FW: BILSEN Paper review

Selam

Sen bu reviewi yaptın mı?

Eray

From: eraytuzun@cs.bilkent.edu.tr <eraytuzun@cs.bilkent.edu.tr>

Date: Sunday, March 24, 2024 at 10:53

To: [REDACTED]

Cc: Selen Uysal <uyselen@gmail.com>, Yahya Elnouby <yelnouby2000@gmail.com>, umutcihan1998@gmail.com <umutcihan1998@gmail.com>

Subject: BILSEN Paper review

Dear All,

Can you please review the following paper ? The paper is about applying H-index in software development

You can access the paper on overleaf by using the link below (please let us know if you cannot access it).

Overleaf Link: <https://www.overleaf.com/project/6414011914cee9db01df4461>



Example of a Reviewer Declining a Review for Personal Reasons





Current Process (As-Is): Step-by-Step Flow

4. **Review Completion**

- Reviewers email authors and coordinator when done.

5. **Feedback on Obtained Reviews**

- After the review deadline, the coordinator manually follows up with authors to assess:
 - Whether all assigned reviewers completed their reviews on time.
 - The perceived quality and usefulness of the received reviews.

Sample Email Sent by a Reviewer After Completing the Review



Emre Dinç

Thu, Jan 22, 3:47 PM (10 days ago)



to eraytuzun@cs.bilkent.edu.tr, İlke, Atakan, mehmet.dedeier@ug.bilkent.edu.tr, me, zehra.iyigun@alumni.bilkent.edu.tr, izgi.tamci@alumni.bilkent.edu.tr, begum.kara@alumni.bilkent.edu.tr, a

Hi all,

I have completed my review. I think the paper introduces a good approach to a time-consuming problem for all software development teams. There are a few points I thought that could be improved, and I added comments about them to Overleaf. To write shortly here, some of them are;

- vague expressions in some places where data could be emphasized more
- turkish file names in the replication package
- not emphasizing significant data enough (such as 83% vs. 57% quiz success from the human-AI or only-AI versions)
- proposing agentic ideas to improve self-reliance of the AI of LACY to capture an understanding of the codebase and business logic

I wish you success and luck in the paper submission process.

Best,
Emre



Current Process (As-Is): Step-by-Step Flow

6. Iteration

- Authors read comments, provide feedback and may ask clarifications on Overleaf comments.
- Reviewers periodically revisit the Overleaf project to respond possible questions and engage in further discussion when needed.
- In cases of rejection or major revision, the process is repeated after manuscript revision. Maintaining a record of previous reviewers is important, as new reviewers are usually assigned in later rounds.

7. Manual Author Checks

- Authors:
 - Use AI-driven review tools for further improvement
 - Check venue requirements and formatting
 - Control a checklist appropriate to the paper type

Sample of a Quality Checklist

Empirical Standards

[About](#) [Standards](#) [Supplements](#) [FAQ](#) [Glossary](#) [People](#)



Reviewer Checklist

[Clear checklist](#)

Essential

yes no

- ☐ ☐ states a purpose, problem, objective, or research question
- ☐ ☐ explains why the purpose, problem, etc. is important (motivation)
- ☐ ☐ defines jargon, acronyms and key concepts
- ☐ ☐ describes the proposed artifact in adequate detail
- ☐ ☐ justifies the need for, usefulness of, or relevance of the proposed artifact
- ☐ ☐ conceptually evaluates the proposed artifact; discusses its strengths, weaknesses and limitations
- ☐ ☐ explains why it is timely to investigate the proposed problem using the proposed method

- ☐ ☐ names the methodology or methodologies used
- ☐ ☐ methodology is appropriate (not necessarily optimal) for stated purpose, problem, etc.
- ☐ ☐ describes in detail what, where, when and how data were collected (see the [Sampling Supplement](#))
- ☐ ☐ describes in detail how the data were analyzed
- ☐ ☐ empirically evaluates the proposed artifact using:
 - [action research](#), in which the researchers intervene in a real organization using the artifact,
 - [a case study](#) in which the researchers observe a real organization using the artifact,
 - [a controlled experiment](#) in which human participants use the artifact,
 - [a quantitative simulation](#) in which the artifact is assessed (usually against a competing artifact) in an artificial environment,
 - [a benchmarking study](#), in which the artifact is assessed using one or more benchmarks, or another method for which a clear and convincing rationale is provided
- ☐ ☐ clearly indicates which of the above empirical methodology is used
- ☐ ☐ EITHER: discusses state-of-art alternatives (and their strengths, weaknesses and limitations)
OR: explains why no state-of-art alternatives exist
OR: provides compelling argument that direct comparisons are impractical
- ☐ ☐ EITHER: empirically compares the artifact to one or more state-of-the-art alternatives
OR: empirically compares the artifact to one or more state-of-the-art benchmarks
OR: provides a clear and convincing rationale for why comparative evaluation is impractical
- ☐ ☐ justifies the selection of the case(s) or site(s) that was(were) studied
- ☐ ☐ describes the site(s) in rich detail
- ☐ ☐ the site is a real laboratory or natural object, not a toy scenario or simulation world



Key Limitations of the Current Process

Manual & Error-Prone Coordination

- Reviewer workload is not balanced.
- Deadlines are tracked manually.
- Missed or late reviews may go unnoticed.
- Does not scale as submissions increase.

No Traceability of Reviewer Performance

- Review quality feedback is: informal, subjective, and not stored.
- No historical data on reviewer performance.



Key Limitations of the Current Process

Heavy Cognitive Load on Authors

- Authors must manually handle:
 - Reviewers' feedbacks
 - AI-based reviews
 - Venue-specific formatting rules (page limits, abstract length, anonymity check)
 - Quality checklist

These tasks are disconnected, undocumented, and error-prone.

No Centralized Visibility or Analytics

- No insight into: reviewer timeliness, workload distribution, and papers previously reviewed by each reviewer.



Why Change Is Needed?

As BILSEN grows:

- Number of papers increases
- Number of reviewers increases
- Number of review rounds increases

Current process does not scale. We need a system that:

- Reduces coordination overhead
- Improves consistency and traceability
- Enables analytics and long-term improvement
- Preserves academic rigor and human control



“To-Be” Automated Bilsen Review Process

Automated Pre-Check & Compliance Validation

Based on the selected venue and track, the system should automatically verify compliance with:

- Page limit
- Abstract word count
- Required sections
- Reference format
- Anonymity requirements by checking for organization and institution anonymization, ensuring PDF metadata (e.g., author and company fields) is cleared, and validating that tool or dataset links do not reveal author identities.
- Detect paper type (research, survey, etc.) and activate the appropriate checklist dynamically.



“To-Be” Automated Bilsen Review Process

Reviewer Assignment & Workload Balancing

- Coordinator selects reviewers & deadline

System:

- Automatically notifies reviewers by email.
- Tracks the assigned reviews per reviewer.
- Enable reviewers to accept or decline review assignments (by providing a valid reason) or request deadline extensions.

Enables:

- Balanced workload distribution.
- Early detection of overload or inactivity.
- Reviewer selection in future review rounds of a paper.



“To-Be” Automated Bilsen Review Process

Automated Coordination & Deadline Management

- Send reminder emails to assigned reviewers as the deadline approaches.
- Alerts to coordinator when deadlines are missed.

Reduces:

- Manual follow-ups.
- Missed or overdue submissions.



“To-Be” Automated Bilsen Review Process

AI-Assisted Review Support (for Authors)

The system supports automated review generation by:

- Highlighting potential issues directly in Overleaf and providing improvement suggestions as free-form comments (ideal one).
- Generating formal review reports based on predefined template.
- Producing annotated PDF documents with highlighted issues and suggested improvements.
- Performing structured, checklist-based compliance checks.

Ensure:

- No hallucinated references
- No unsupported claims



“To-Be” Automated Bilsen Review Process

Reviewer Evaluation Mechanism

After review completion, authors rate reviewers on:

- Quality of comments
- Quantity of feedbacks
- Timeliness

Ratings contribute to:

- Reviewer performance statistics
- Long-term quality monitoring



“To-Be” Automated Bilsen Review Process

Reviewer Evaluation Mechanism

Dashboard should show:

- Ranked list of reviewers based on performance metrics (review quality, quantity, and timeliness).
- Assigned and completed reviews per reviewer, tracked monthly, yearly, and overall.
- Review assignment acceptance and rejection rates per reviewer, tracked monthly, yearly, and overall.
- Percentage of on-time versus delayed reviews per reviewer, tracked monthly, yearly, and overall.
- Average reviewer score.

Purpose

- Enable data-driven decisions.
- Monitor health of the review process.



“To-Be” Automated Bilsen Review Process

Venue Recommendation & Submission Support (Optional)

Recommend suitable venues based on:

- Paper type
- Content
- Allow authors to select a target venue

Enable direct submission once:

- All checks are satisfied



Thanks for your Attention

Questions? Feel free to reach out via Slack (“Fereshteh Vedadi”).